

1. Hyperbolic functions

What students need to learn:

Definition of the six hyperbolic functions in terms of exponentials. Graphs and properties of the hyperbolic functions.

For example, $\cosh x = \frac{1}{2}(e^x + e^{-x})$, $\operatorname{sech} x = \frac{1}{\cosh x} = \frac{2}{e^x + e^{-x}}$.

Students should be able to derive and use simple identities such as $\cosh^2 x - \sinh^2 x \equiv 1$ and $\cosh^2 x + \sinh^2 x \equiv \cosh 2x$ and to solve equations such as $a \cosh x + b \sinh x = c$.

Inverse hyperbolic functions, their graphs, properties and logarithmic equivalents.

E.g. $\operatorname{arsinh} x = \ln[x + \sqrt{1 + x^2}]$. Students may be required to prove this and similar results.
